GONGFAN CHEN

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Dept. of Engineering Technology and Construction Management
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EDUCATION

- **North Carolina State University**, Raleigh, NC, U.S.
  Ph.D. in Civil Engineering (4.0/4.0) Aug 2019 – Aug 2023
  M.S. in Electrical and Computer Engineering (4.0/4.0) Aug 2021 – May 2023
- **University of Michigan**, Ann Arbor, MI, U.S.
  M.S. in Civil Engineering (3.7/4.0) Aug 2017 – May 2019
- **North China Electric Power University**, Beijing, China
  B.S. in Engineering and Management (87/100) Sept 2013 – June 2017

PROFESSIONAL APPOINTMENTS

- **Assistant Professor,** University of North Carolina at Charlotte, August 2025 – present
- **Affiliated Faculty,** Charlotte AI Institute (CLTAI^2), University of North Carolina at Charlotte, Sept. 2025 – present
- **Affiliated Faculty,** Charlotte AIR Institute, University of North Carolina at Charlotte, Sept. 2025 – present
- **Innovation Incubator Faculty,** urbanCORE, University of North Carolina at Charlotte, Oct. 2025 – present
- **Visiting Research Scholar,** North Carolina State University, August 2025 – present
- **Postdoctoral Research Scholar,** North Carolina State University, August 2023 – August 2025
- **Graduate Research Assistant,** North Carolina State University, August 2019 – August 2023
- **Graduate Teaching Assistant,** North Carolina State University, August 2019 – August 2023
- **Graduate Teaching Assistant,** University of Michigan, August 2018 – December 2018
- **BIM Engineer,** China Railway 24th Bureau Group Co., Ltd., May 2018 – August 2018
- **BIM Lecture,** North China Electric Power University (Beijing), January 2016 – June 2016
- **Research Assistant,** North China Electric Power University (Beijing), September 2014 – June 2017

PUBLICATIONS

Book Chapters:

- **[B1]**. Chen, G., Ovid, A., Alsharef, A., Pedraza, M., Albert, A., and Jaselskis, E. (2025). "Wearable Edge AI Application for Real-Time Construction Safety Monitoring" *AI, Robotics, and Automation in Construction – Emerging Technologies for Sustainability, Efficiency, and Safety,* Springer Nature (submitted).

Referred Journal Articles:

- **[J11]**. Chen, G., Pedraza, M., Albaloul, R., Albert, A., and Jaselskis, E. (2026). "Advancing Fit-for-Purpose Access to CII Best Practices Using Multimodal Large Language Models." *Journal of Management in Engineering* (under revision).
- **[J10]**. Chen, G., Jaselskis, E., Tamer, A., and Folz, A. (2025). "Using an LLM-Powered Framework for Automatic Risk Identification and Mitigation from Past Claims and Supplemental Agreements." *Journal of Computing in Civil Engineering,* 40 (2): 04025151.
- **[J9]**. Diaz, B., Chen, G., Delgado, C., and Jaselskis, E. (2025). "Supporting Generative AI Literacy: Exploring the Pedagogical Roles Students Assign ChatGPT and Impact on Course Grades." Comunicar, 33(82), 46–61.
- **[J8]**. Shafaay, M., Alqahtani, F. K., Alsharef, A., and Chen, G. (2025). Modeling construction cost overrun risks at the FEED stage for mining projects using PLS-SEM. *Journal of Asian Architecture and Building Engineering*, 1–17.
- **[J7]**. Chen, G., Alsharef, A., Ovid, A., Albert, A. and Jaselskis, E. (2025). "Meet2Mitigate: An LLM-powered framework for real-time issue identification and mitigation from construction meeting discourse." *Advanced*

Engineering Informatics, 64: 103068.

- **[J6]**. Chen, G., Alsharef, A., and Jaselskis, E. (2024). "Construction Jobsite Image Classification Using an Edge Computing Framework." *Sensors,* 24 (20):6603.
- **[J5]**. Chen, G., M. Liu, H. Li, S. M. Hsiang, and A. Jarvamard. (2023). "Motivating Reliable Collaboration for Modular Construction – A Shapley Value-Based Smart Contract." *Journal of Management in Engineering*, 39 (6): 04023042. (Editor's Choice)
- **[J4]**. Chen, G., M. Liu, Y. Zhang, Z. Wang, S. M. Hsiang, and C. He. (2023). "Using Images to Detect, Plan, Analyze, and Coordinate a Smart Contract in Construction." *Journal of Management in Engineering*, 39 (2): 1–18.
- **[J3]**. He, C., M. Liu, Y. Zhang, Z. Wang, S. M. Hsiang, G. Chen, W. Li, and G. Dai. (2023). "Space-Time-Workforce Visualization and Conditional Capacity Synthesis in Uncertainty." *Journal of Management in Engineering*, 39 (2): 04022071.
- **[J2]**. Chen, G., H. Li, M. Liu, S. M. Hsiang, and A. Javanmardi. (2022). "Knowing what is going on – a smart contract for modular construction." *Canadian Journal of Civil Engineering*, 50 (3): 210-223. (Invited paper).
- **[J1]**. He, C., M. Liu, Y. Zhang, Z. Wang, S. M. Hsiang, G. Chen, and J. Chen. (2022). "Exploit Social Distancing in Construction Scheduling: Visualize and Optimize Space–Time–Workforce Tradeoff." *Journal of Management in Engineering*, 38 (4): 1–15.

Referred Conference Articles:

- **[C19]**. D'Souza, K.., Chen, Y., and Chen, G. (2026). "Computational Mapping of Occupational Risk from Online Narratives: An NLP Approach to Worker Safety in Construction." *43rd International Symposium on Automation and Robotics in Construction (ISARC) 2026* (under preparation).
- **[C18]**. Chen, Y.., Chen, G., Chen, D., and Song, L. (2026). "Wearable Edge-AI Assistant for Multilingual Safety Communication on Construction Sites: A Heat Stress Prevention Application." *43rd International Symposium on Automation and Robotics in Construction (ISARC) 2026* (under preparation).
- **[C17]**. Jaselskis, E., Chen, G., Pedraza, M., Albaloul, R., Albert, A., and Mostafa, M. (2026). "Can AI improve construction project performance?" *The Sixth European Structural Engineering and Construction Conference2026* (submitted).
- **[C16]**. Albaloul, R., Chen, G., and Jaselskis, E. (2026). "An AI-Enabled Knowledge Curation Framework for Actionable Best Practices in Nuclear Power Construction Projects." *International Conference on Computing in Civil and Building Engineering (ICCCBE) 2026* (under revision).
- **[C15]**. Chen, Y.., and Chen, G. (2026). "Exploring Public Perceptions of U.S. Highway Pedestrian Presence via Reddit and NLP." *ASCE International Conference on Transportation & Development 2026.*
- **[C14]**. Banerjee, S., Chen, G., Jaselskis, E., and Cho, J. (2026). "Exploring Public Perceptions of COVID-19's Impact on the Construction Sector Using Twitter-Based Text Analytics." *Construction Research Congress 2026.*
- **[C13]**. Ovid, A., Albert, A., Uddin, S., Alsharef, A., and Chen, G. (2026). "Application of Neural Networks in BIM and 3D Reconstruction: A Systematic and Scientometric Review." *Construction Research Congress 2026.*
- **[C12]**. Ovid, A., Albert, A., Chen, G., Alsharef, A., and Uddin, S. (2026). "Evaluating Closed-Set and Open-Vocabulary Object detectors in Real-Time Construction Environment." *Construction Research Congress 2026.*
- **[C11]**. Ovid, A., Albert, A., Alsharef, A., Uddin, S., and Chen, G. (2026). "Automated Classification of Construction Incident Reports: A Systematic Review and Evaluation of Models." *Construction Research Congress 2026.*
- **[C10]**. Chen, G., Li, Y., Jaselskis, E., and Liu, Y. (2026). "Portable Lightweight Real-Time AI System Using Edge-RAG for Construction Best Practice Interactions." *Construction Research Congress 2026.*
- **[C9]**. Chen, G., Banerjee, S., Ye, X., Liu, L., and Chung, I. (2026). "An LLM-Powered Agentic Framework for Automatically Converting Construction Contracts into Smart Contracts." *Construction Research Congress 2026.*
- **[C8]**. Chen, G., Pedraza, M., Albaloul, R., Albert, A., and Jaselskis, E. (2025). "Developing a Fit-for-Purpose Best Practice Knowledge Handbook using Generative AI." *CSCE Construction Specialty Conference / ASCE Construction Research Congress (CRC) 2025.*
- **[C7]**. Chen, G., and Liu, Y. (2025). "Leveraging Large Language Models for Voice-Activated Robotic Teleoperation in Construction." *ASCE International Conference on Computing in Civil Engineering 2025.*

- **[C6]**. Mapare, V., Liu, Y., Zheng, D., and Chen, G. (2025). "Enhancing the Worker-Robot Interaction for Construction Task Operation: A Preliminary Study Using Large Language Model-based Methods." *ASCE International Conference on Computing in Civil Engineering 2025*.
- **[C5]**. Chen, G., Alsharef, A., and Jaselskis, E. (2024). "A Novel Edge Computing Framework for Construction Nail Detection under Conditions of Constrained Computing Resources." *ASCE International Conference on Computing in Civil Engineering 2024, Pittsburgh, Pennsylvania, 2024*.
- **[C4]**. Chen, G., Alsharef, A., and Jaselskis, E. (2024). "Integrating Sensor-Empowered Federated Learning and Smart Contracts for Automatic Sustainable Infrastructure Management." *ASCE International Conference on Computing in Civil Engineering 2024, Pittsburgh, Pennsylvania, 2024*.
- **[C3]**. Chen, G., C. He, S. Hsiang, M. Liu, and H. Li. (2023). "A mechanism for smart contracts to mediate production bottlenecks under constraints." *31st Annual Conference of the International Group for Lean Construction (IGLC)*. Lille, France.
- **[C2]**. Li, H., G. Chen, M. Liu, S. M. Hsiang, and A. Jarvamardi. (2023). "Situation awareness based smart contract for modular construction." *Proceedings of the Canadian Society of Civil Engineering Annual Conference 2021*, 363–373. Singapore: Springer Nature Singapore (Best Student Paper for Awardee: Gongfan Chen).
- **[C1]**. He, C., M. Liu, Z. Wang, G. Chen, Y. Zhang, and S. M. Hsiang. (2022). "Facilitating Smart Contract in Project Scheduling under Uncertainty—A Choquet Integral Approach." *Construction Research Congress 2022*, 930–939.

GRANTS

- **PI, Investigating Edge AI Applications to Support Informed Construction Jobsite Decision-Making**, Office of Undergraduate Research (OUR)-Spring 2026 OUR Scholars Mentored Project. UNC Charlotte, $2,000.
- **PI, Investigating the Role of Multimodal AI to Assist Engineering Plan Reading in Undergraduate Education**, Scholarship of Teaching and Learning (SOTL) Grant, UNC Charlotte, $6,100
- **Co-PI, Evaluating the Accuracy and Effectiveness of AI-Mediated Information Transformation in STEM: From Lecture Slides to Instructional Video,** Scholarship of Teaching and Learning (SOTL) Grant, UNC Charlotte, $6,000

THESIS COMMITTEE

Master Student

- **Jiaqi Sun**, Data Science and Business Analytics, University of North Carolina at Charlotte (S26)

TEACHING AND MENTORING EXPERIENCE

Instructor

- **CMET2105 Plan Reading**, University of North Carolina at Charlotte (F25, S26)

Teaching Assistant:

- **North Carolina State University**
- CE365 Construction Equipment and Method (S21); CE463 Construction Estimating, Planning, and Control (F19, 20, 21); CE469 Construction Engineering Project (S2, 22, 23); CE564 Legal Aspects of Contracting (S22, 23, 24); CE592 CII Best Practices (F21); CE592 Global Construction Practices (F23)
- **University of Michigan**
- CEE431 Construction Contracting (F18)

Mentees

- **Graduate Student**
  - Michael Pedraza (Civil, NCSU); Roya Albaloul (Civil, NCSU); Myat Mon Aye (Civil, NCSU)
- **Undergraduate Student**

- Saniya Pritchett (Construction, UNC Charlotte) Yongxuan Zhang (Mathematics, NCSU); Shilun Zhao (ME, NCSU); Hongxu Liu (Engineering, NCEPU); David Daildly (Construction, NCSU); Sidney Perkinson (Construction, NCSU); Ruby Cain (Construction, NCSU); Austin Booth (Construction, NCSU)

AWARDS

- **Travel Grants Award**, NSF Workshop at I3CE2024 Conference, 2024
- **ASCE Journal of Management Editor's Choice**, North Carolina State University, Raleigh, 2023
- **People's Choice Award**, NCSU CCEE 3-Minute Thesis Competition, 2022
- **Best Student Paper Award**, Canadian Society of Civil Engineering Annual Conference, 2021
- **Graduate Merit Award (GMA) Fellowship**, North Carolina State University, Raleigh, 2019
- **Second Place Award**, "Rush Hour" Business Decision-Making and Management Competition, Role: team captain, Beijing, China, 2017
- **Exchange Scholarships**, National Chi Nan University, Puli, Taiwan, 2015
- **First Place Award**, National Business Road Management Decision-making Competition, Role: team Captain, Shanghai, China, 2015
- **Department Outstanding Student**, North China Electric Power University (NCEPU), Beijing, China, 2014 – 2017

CERTIFICATE

- **Construction Industry Institute (CII) Professional Development Certificate**, Nashville, TN, 2025
- **PL-300 Microsoft Power BI Data Analyst Certification Training Program**, Raleigh, NC, 2025
- **Generative AI for Business with Microsoft Azure OpenAI Program**, Raleigh, NC, 2024
- **IBM Data Science** Professional Certificate, 2022
- **BIM Engineering Modeling Technical Certificate**, Certificate No. 160090200652, Beijing, China, 2016
- **BIM Engineer (Modeler) Professional License,** Certificate No. BJLJMBIMGCS00690, Beijing, China, 2016

PRESENTATION

Invited Lectures:

- **Blockchain-Enabled Smart Contract for the Construction Industry**, CE 4130 Construction Contracts, California State Polytechnic University-Pomona, 2025
- **Leveraging BERTopic Modeling to Identify Risk Categories for Claims and Supplementary Agreements**, CE514- Decision Making and Risk Management in Construction, King Saud University, Saudi Arabia, 2024

Poster Presentation:

- **Meet2Mitigate: An LLM-Powered Framework for Real-Time Issue Identification and Mitigation from Construction Meeting Discourse**, Applied AI in Engineering & Computer Science Symposium, North Carolina State University, 2024

Conference Presentation:

- **Enabling Real-Time AI Applications in Construction Site Safety: A Novel Edge Computing Approach for Nails Detection Under Resource-Constrained Environment**, ASCE International Conference on Computing in Civil Engineering, Pittsburgh, Pennsylvania, 2024
- **Integrating Sensor-Empowered Federated Learning and Smart Contracts for Automatic Sustainable Infrastructure Management**, ASCE International Conference on Computing in Civil Engineering, Pittsburgh, Pennsylvania, 2024
- **A Fit-for-Purpose (F4P) Handbook for Effective Implementation of CII Best Practices**, Construction Research Congress 2024, Des Moines, Iowa, 2024
- **Insights for Effective Risk Management in Transportation Projects**, Transportation Research Board (TRB)

Annual Meeting, Washington DC, 2024

- **Situation Awareness Based Smart Contract for Modular Construction**, Canadian Society of Civil Engineering Annual Conference, Niagara Falls, Ontario (Online), 2021

University Competition:

- **Situation Awareness Smart Contracts for Construction Management**, CCEE 5th Annual Three Minute Thesis Competition, North Carolina State University, 2022

REVIEW SERVICE

- Advanced Engineering Informatics, 2025 – Present (#: 3)
- ASCE Journal of Construction Engineering and Management, 2025 – Present (#: 3)
- International Journal of Construction Management, 2025 – Present (#: 5)
- Automation in Construction, 2025 – Present (#: 14)
- ASCE Journal of Computing in Civil Engineering, 2024 – Present (#: 16)
- Results in Engineering, 2024 – Present (#: 3)
- ASCE Journal of Management in Engineering, 2023 – Present (#: 7)
- Construction Innovation, 2022 – Present (#: 26)
- Sustainability, 2024 – Present (#: 2)
- Buildings, 2024 – Present (#: 1)
- Big Data and Cognitive Computing, 2024 – Present (#: 2)
- Applied Science, 2024 – Present (#: 2)
- Intelligent Infrastructure and Construction, 2024 – Present (# : 1)
- Peer J Computer Science, 2024 – Present (#: 1)
- Discover Artificial Intelligence, 2024 – Present (#: 1)
- Peer-to-Peer Networking and Application, 2024 – Present (#: 2)
- ASCE CI&CRC Joint Conference, San Antonio, Texas, 2026 (#: 8)
- ASCE International Conference on Computing in Civil Engineering, Pittsburgh, Pennsylvania, 2025 (#: 5)
- ASCE International Conference on Computing in Civil Engineering, Pittsburgh, Pennsylvania, 2024 (#: 5)
- International Symposium on Automation and Robotics in Construction (ISARC), Dubai, UAE, 2021 (#: 1)

PROFESSIONAL SERVICE

- **Associate Member**, American Society of Civil Engineering (ASCE)
- **Member**, Digital Project Delivery, ASCE
- **Younger Member**, Data Sensing & Analytics Committee, ASCE
- **Corresponding Member**, Infrastructure Systems Committee, ASCE

SKILLS

- **Language Proficiency:** Professional English, Native Chinese
- **Programming Languages:** Python, JavaScript, SQL, MATLAB, R, SAS, Julia, Solidity
- **Tools/Software:** Power BI, Revit, Navisworks, BIM 360, HoloLens, Unity 3D, AutoCAD, Synchro, Lumion, GIS, Bentley, Primavera, Bluebeam Revu, Ganache, MetaMask, TensorFlow, PyTorch, Arena, Stroboscope
- **Hardware:** Raspberry Pi, Nvidia Jetson, wearable sensors, Edge TPU